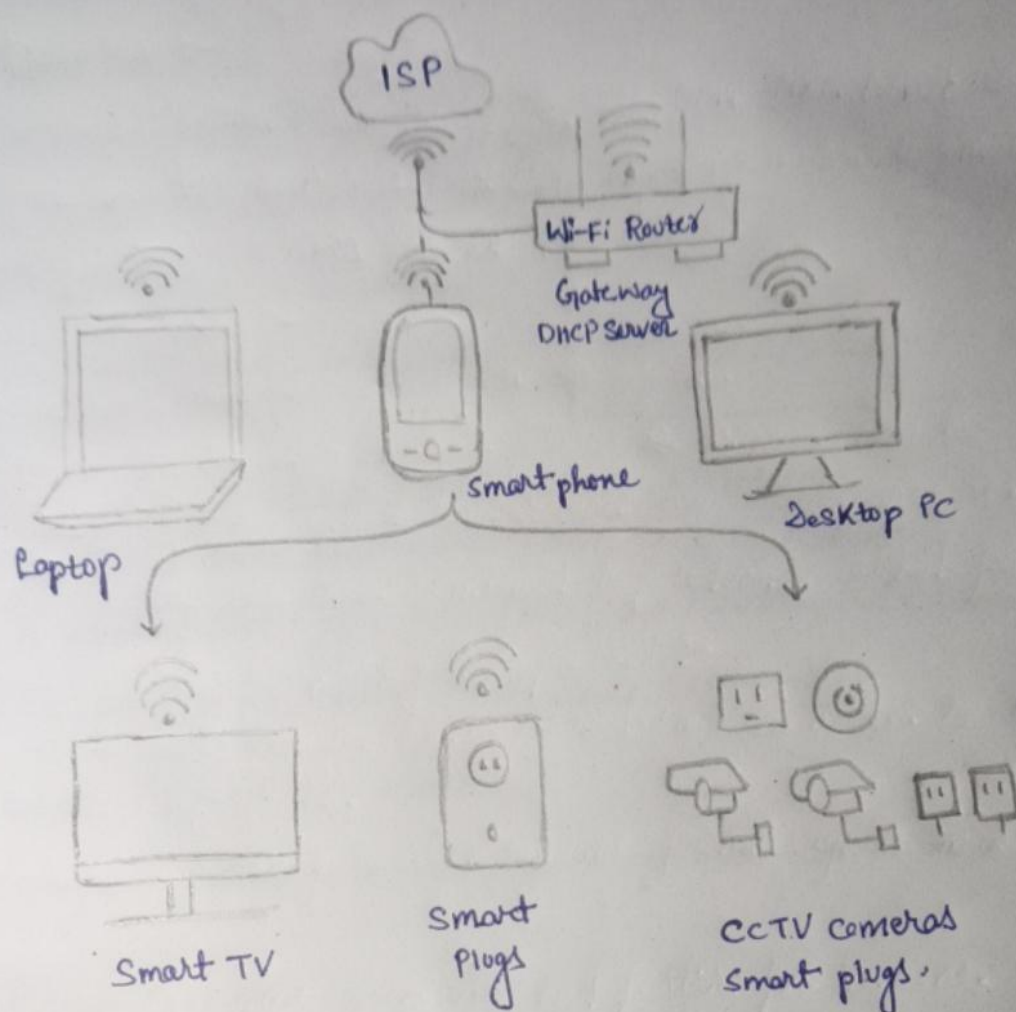


Assignment-1 (CA-Network)



Home Network Topology.

1. ISP (Internet service provider)
 - provides internet connection via DSL, fiber, or cable.
 - Connects to the modem or modem-router combo.
2. Router/Wifi Access point
 - central hub for all devices.
 - Assigns IP addresses via DHCP.
 - Acts as NAT gateway (Network address Translation).
 - Support both wired (Ethernet) & wireless (Wi-Fi) connection.

3. Devices

- Laptop / phone / Smart TV / IoT : Connected via Wi-Fi.
- Desktop PC : often connected via Ethernet for stable and faster connection.

4. Internet Network (LAN)

- All devices share a private IP range (e.g., 192.168.0.x).
- Devices can communicate with each other (e.g. file sharing, printer access).